12.1.2 Mathematics in Economics



Sir William Petty
1623-1687

Sir William Petty declared that he wanted to reduce political and economic matters in terms of number, weight and measure. He was the first one to use mathematics in economics. The first known writer to apply mathematical method to economic problems was Giovanni Ceva (1711), an Italian.

12.1.3 Uses of Mathematical Methods in Economics

1. Mathematical Methods help to present the economic problems in a more precise form.
2. Mathematical Methods help to explain economic concepts.
3. Mathematical Methods help to use a large number of variables in economic analyses.
4. Mathematical Methods help to quantify the impact or effect of any economic activity implemented by Government or anybody. There are of course many other uses.



Think and Do

- Who is the father of Economics? Did he use any of the mathematical tools in his contributions? If yes, list out.
- Find out the mathematical tools, which are used by you in your daily routine life.

12.2 Functions

12.2.1 Definition

A function is a mathematical relationship in which the values of a dependent variable are determined by the values of one or more independent variables.

Functions with a single independent variable are called Simple Univariate functions. There is a one to one correspondence. Functions,with more than one independent variable, are called Multivariate functions. The independent variable is often designated by X. The dependent variable is often designated by Y. For example, Y is function of X which means Y depends on X or the value of Y is determined by the value of X. Mathematically one can write $Y = f(X)$.

12.2.2 Linear Equation

A statement of relationship between two quantities is called an equation. In an equation, if the largest power of the independent variable is one, then it is called as Linear Equation. Such equations when graphed give straight lines. For example $Y = 100–10X$.

For a straight line, there are two variables namely X and Y. X is called independent variable and Y is called dependent variable.

When 'X' value increases by one unit, then the corresponding change in the 'Y' value is called as the slope of the line. Slope of the line is obtained by the formula,

m = slope (marginal change)

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